

Vilnius University Šiauliai Academy



Object Oriented Programming
Practical Work 6:
Abstract classes, interfaces, trait, namespaces

Lecturer: Dr. Donatas Dervinis
Student: Nikola Odjakov
GitHub Repository: [Project](#)

Šiauliai, 2026

Test Report

Overview

This document summarizes the testing of a simple User Management System built with PHP and MySQL.

The system allows users to register, log in, view profiles, and (if admin) manage other users.

Test environment

PHP version: 8.5.6;

Database: MySQL (localhost), through Laragon, used HeidiSQL;

Operating System: Windows 10;

Browser used: Mozilla Firefox.

Tests Conducted

To test out the program, multiple features were tested to see whether they function correctly. The features are as follows:

- Accessing the web page;
- User Registration;
- Session handling, where the program knows whether a user is logged in or not, and what kind of user.
- Login, and logout;
- Viewing the user profile list;
- Deleting users through the admin user;
- Whether data is successfully encrypted, and decrypted, and whether passwords remain hashed.

Summarized Test Results:

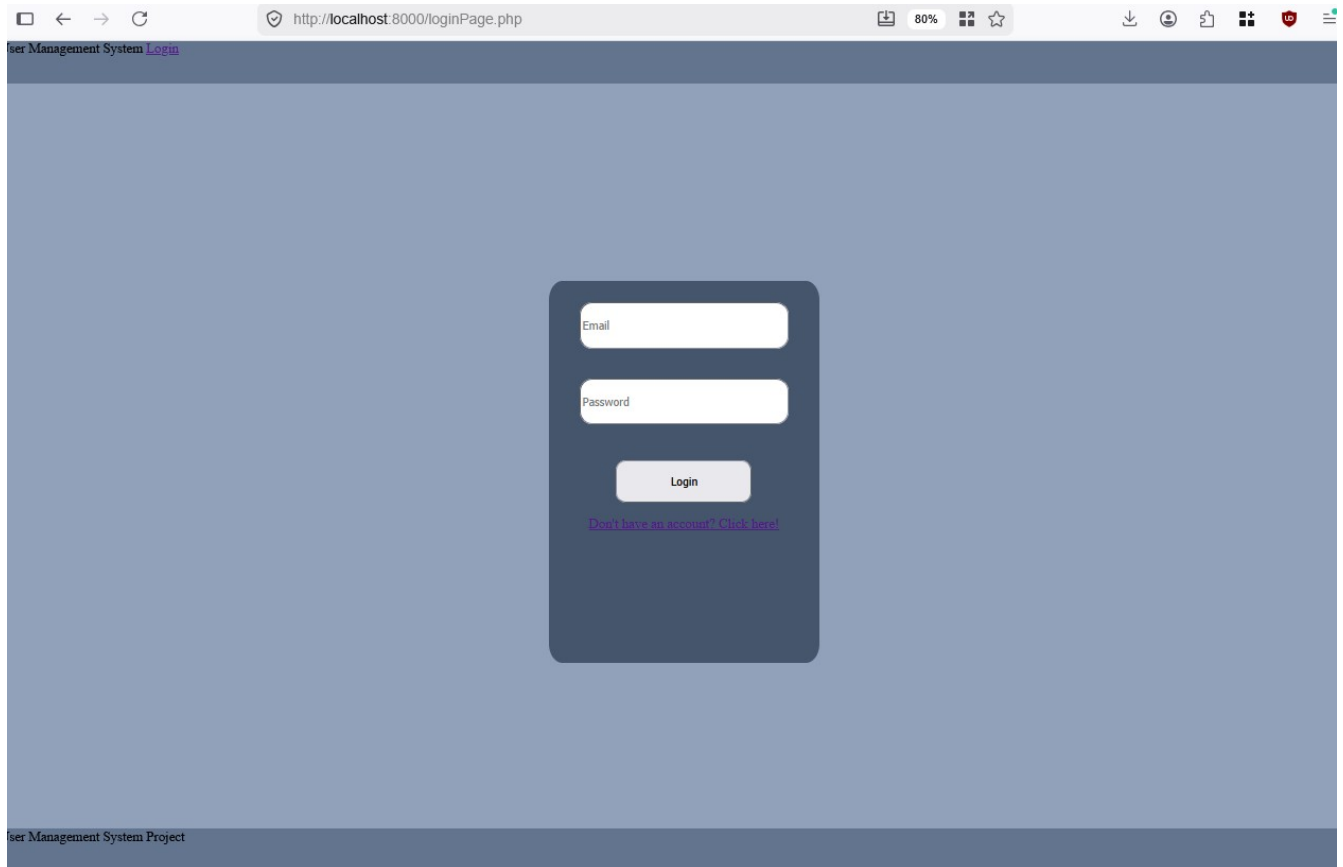
Feature	What should happen	(if not passed)	Status
Page access	The default localhost:8000 redirects to the homepage	/	Passed
Registration	Successful user registration	/	Passed
Login	User logs in successfully	/	Passed
Login (Admin)	Admin logs in successfully	/	Passed
Logout	User is logged out	/	Passed
Session Handling	Whether the program correctly detects the user status	/	Passed
User Profile	The details of the user logged in should be shown	/	Passed
User list (as User)	Shows all users	/	Passed
User List (as Admin)	Shows all users, with the delete functionality visible	/	Passed
Delete user	User is removed from DB	/	Passed
Access control	Non-admin cannot delete users	/	Passed
Encryption	All data is properly encrypted and hashed	/	Passed
Logging	Each user action is correctly logged	Despite multiple different attempts, in the present iteration, logging is not possible	Failed

- Overall, 10 out of 11 tests were successful. The Logging feature will be corrected in subsequent iterations.

Test Demonstrations:

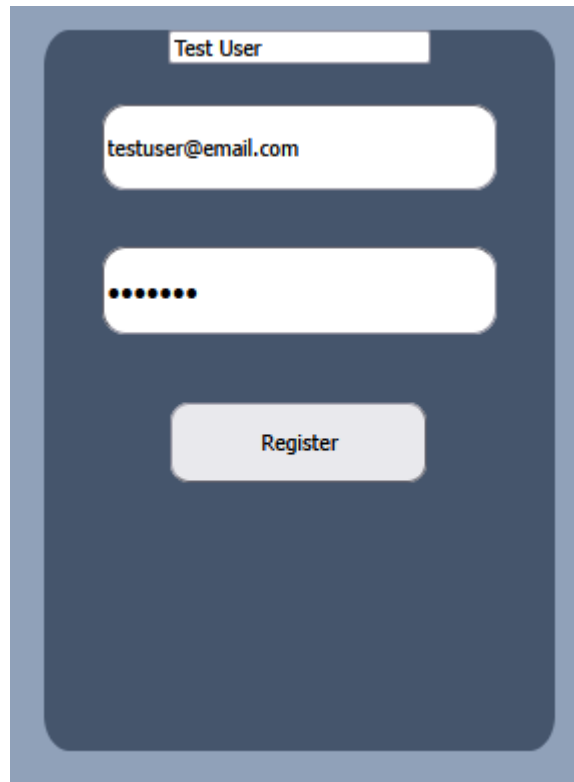
Page Access:

Upon accessing <http://localhost:8000>, the user is automatically redirected to the login page, due to the homepage checking for user status, and in case it detects a user is not logged in, it then redirects to the login page.



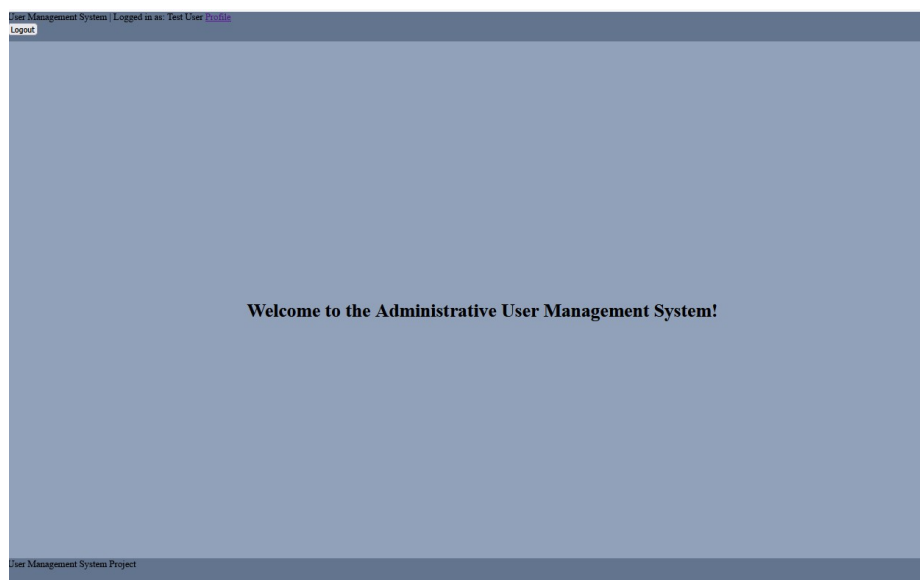
User Registration:

-Upon clicking the hyperlink for registration, the user is navigated to the registration page. The user is prompted with multiple submission fields. Here, a test user was created.

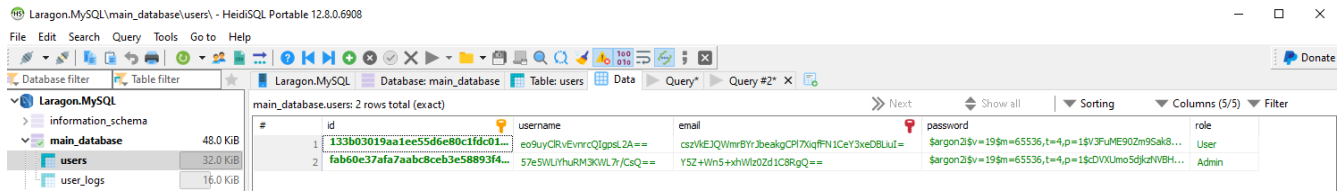


A user registration form with a dark blue background and rounded corners. At the top, there is a text input field containing "Test User". Below it is an email input field containing "testuser@email.com". Underneath the email field is a password input field with masked characters ".....". At the bottom of the form is a light blue "Register" button.

- Upon registration the user is brought to the homepage:



- With some session handling, we can see that the user was logged in in the header, alongside the Profile and Logout options appearing. To confirm that the user was added to the database, here is the database list:



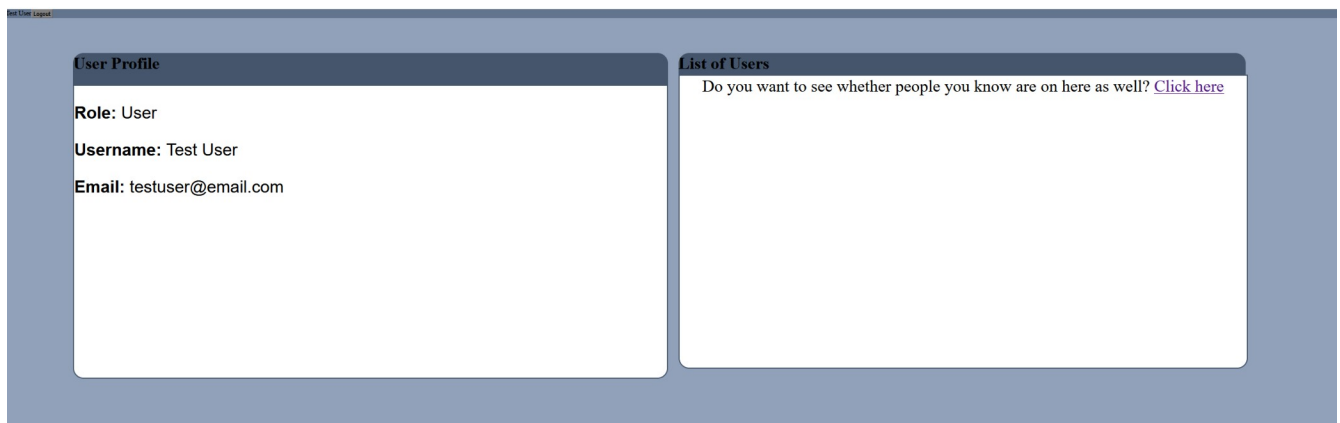
The screenshot shows the HeidiSQL interface with the 'users' table selected. The table contains two rows of data, with the first row highlighted. The columns are: #, id, username, email, password, and role.

#	id	username	email	password	role
1	133b03019aa1ee55d6e80c1fd01...	eo9uyCRvEvmrcQjgsL2A==	cszVKEJQWmr8Y1JbeaigCP7XiqfN1CeY3xeDBLuL=	\$argon2id\$v=19\$m=65536,t=4,p=19V3FuME902m9Sak8...	User
2	fab60e37afa7aabc8ceb3e58893f4...	57e5WLiYruRM3KWL7r/CsQ==	Y5Z+Wn5+XhWlzo2d1C8RgQ==	\$argon2id\$v=19\$m=65536,t=4,p=19cDVXUmo5djzNVBH...	Admin

-At ID1, We can see that a new user was added. For sake of encryption purposes the id, username, and email are encrypted, and the password is hashed. This confirms that both the data is successfully encrypted, as well as successfully added to the database.

User Profile:

Upon clicking the hyperlink to “user profile” from the header, the user is navigated to the UsersProfilePage.php:



The screenshot shows the 'User Profile' page. It displays the following information:

- Role: User
- Username: Test User
- Email: testuser@email.com

On the right, there is a 'List of Users' section with a link: 'Do you want to see whether people you know are on here as well? [Click here](#)'.

-As shown, the page successfully returns the logged in user’s role, username and email.

List of Users:

Upon clicking on the option to see the list of users, the user is prompted to a new page:



The screenshot shows the 'List of Users' page. It displays a list of users:

Test User
Admin

-Where both the logged in user, and the administrative user are listed. For test purposes, Additional users were created to confirm whether user types are globally shown within the list to other users.

While logged in as another user, and following the same user flow, it can be seen that all registered users are listed:



User logout:

Upon clicking the “Logout” button in the header, the user is logged out and brought back to the homepage:



Admin User:

Login:

For security purposes, administrative users cannot be registered through the registration page, instead, in this circumstance the administrative user is hard-coded.

```
<?php
require_once __DIR__ . "/autoloader.php";
require_once __DIR__ . "/database.php";

$stmt = $pdo->prepare("SELECT COUNT(*) FROM users WHERE role = 'Admin'");
$stmt->execute();
$encryption = new \App\Security\Encryption();
if ($stmt->fetchColumn() == 0) {

    $stmt = $pdo->prepare("
        INSERT INTO users (id, username, email, password, role)
        VALUES (:id, :username, :email, :password, :role)
    ");

    $stmt->execute([

        ":id" => bin2hex(random_bytes(16)),
        ":username" => $encryption->encrypt("Admin"),
        ":email" => $encryption->encrypt("admin@email.com"),
        ":password" => password_hash("adminStrongPassword", PASSWORD_ARGON2I),
        ":role" => "Admin"
    ]);
}
```

-For the first attempt during this test, logging in with the admin details was unsuccessful due to a plain text email being compared to the stored encrypted value. Upon fixing this, we are navigated to the following user profile page:

[Admin Logout](#)

User Profile

Role: Admin

Username: Admin

Email: admin@email.com

List of Users

Do you want to see whether people you know are on here as well? [Click here](#)

User List Admin:

Upon entering the user list as a logged in admin user we are prompted with the following page:



This confirms both the functionality of the page as well as the session handling where the delete privilege is only granted to administrative users.

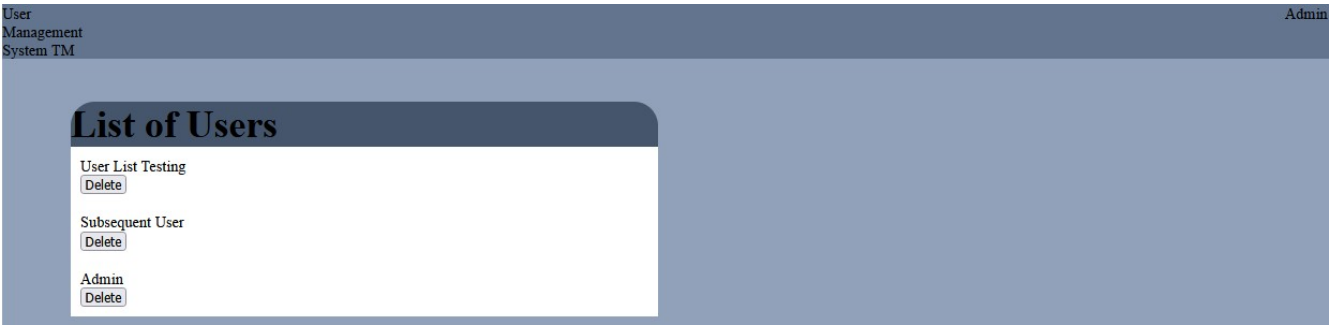
User Deletion:

Present table state:

#	id	username	email	password	role
1	133b03019aa1ee55d6e80c1fdc01...	eo9uyCIRvEvmrcQlgsL2A==	cszVkJEQWmrBYrJbeakgCPI7XiqfFN1CeY3xeDBLuI=	\$argon2\$v=19\$m=65536,t=4,p=1\$V3FuME90Zm9Sak8...	User
2	86c5a7580ea51c8bdc4e9cc6ab4a...	UTVkfGt8ksGZAU4QWUwTmou76ja+RyNO99rXJ/ITIb8I=	TtVv9ZVXWk+RUqOtv7aGPeAHDb4eORqzghVEIm6yq8=	\$argon2\$v=19\$m=65536,t=4,p=1\$SmZzWUpwMzZouE...	User
3	e8745ede3fe39a053d7e944f8bf3...	RSv7Vfa06CXp8/g9A2p0GQ==	SQJ6EVZSBjnyxxx7V0gaCuMUJAHW8Kx6va688g4AXc=	\$argon2\$v=19\$m=65536,t=4,p=1\$T0JwT12mNqpbGQ...	User
4	ec69f41aa8fc1f947d69def947814...	57e5WLYhuRM3KWL7r/CsQ==	YSZ+Wn5+xhWfz0Zd1C8RgQ==	\$argon2\$v=19\$m=65536,t=4,p=1\$SN4OFRTa2FpNFRJ...	Admin

Upon pressing the “Delete” button for “Test User”

The page updates to the following:



Thus Removing the Test User from the user list.

To confirm whether this occurred, the table is checked once more:

	id	username	email	password	role
1	86c5a7580ea51c8bdc4e9cc6ab4a...	UTVKFdT8ksGZAU4QWUwTmou76ja+RyNO99rX/JTib8t=	TtVy9ZVXWk+RUqOtv7aGPeAHIdb4eORqzghVEIm6yq8=	\$argon2i\$y=19\$m=65536,t=4,p=1\$SmZzWUpwMzZoUE...	User
2	e8745ede3fe39a053d7e944f8bf3...	RSv7Yfa06CXp8/g9A2p0GQ==	SQJi6EVZSBjhyxxx7rV0gacuMUAHw8Kx6va688g4AXc=	\$argon2i\$y=19\$m=65536,t=4,p=1\$T0JwT12mNkplbGQ...	User
3	ec69f41aa8fc1f947d69def947814...	57eSWUyfuRM3KWL7r/CsQ==	YSZ+Wn5+xbWlZ0Zd1C8RgQ==	\$argon2i\$y=19\$m=65536,t=4,p=1\$SIN4OFRTaZFpNFRI...	Admin

Comparing the user ID's, we can infer that Test User was successfully deleted due to the disappearance of the 133b-x (#1) Id.

Logging:

The logging functionality successfully ran prior to the session implementation. It was partially integrated, but is not operational in displaying the log data in the latest version, as the output is not displayed.

Short conclusion

The system works as expected for the implemented features, and most tests were successful. Some issues were found during testing, which were resolved. Logging functionality was the only part that did not fully work in the final version, although it functioned during earlier development. The Auth Interface is implemented, but authentication logic was mainly handled by the AuthService as the project evolved. Overall, the project demonstrates use of PHP sessions, OOP principles, encryption, database integration, and was a great learning experience for learning PHP.

References:

Microsoft. (n.d.). Visual Studio Code [Computer Software]. From <https://code.visualstudio.com>
Laragon. (n.d.). *Laragon* [Computer software]. <https://laragon.org>